Descriptive Statistics

GENERAL ASSEMBLY

Learning Objectives

By the end of this lesson, you should be able to:

- Define “statistic.”
- Describe how descriptive statistics relate to the practice of data science.
- Understand and organize the following terms: quantitative, qualitative, ordinal, unordered, datetime, discrete, continuous.
- Understand summation notation.
- Define, compare, and contrast measures of central tendency.
- Describe sensitivity and robustness of descriptive statistics.
- Define, compare, and contrast measure of spread.
- Derive standard deviation as a measure of spread.
- Interpret correlation as a measure of the strength of the linear relationship between two quantitative variables.

Descriptive Statistics

- Descriptive statistics **summarize one or more variables** using a few numbers.
  - A **statistic** is a value computed from our data.

- This is often *really* important for communicating with other people about your work.

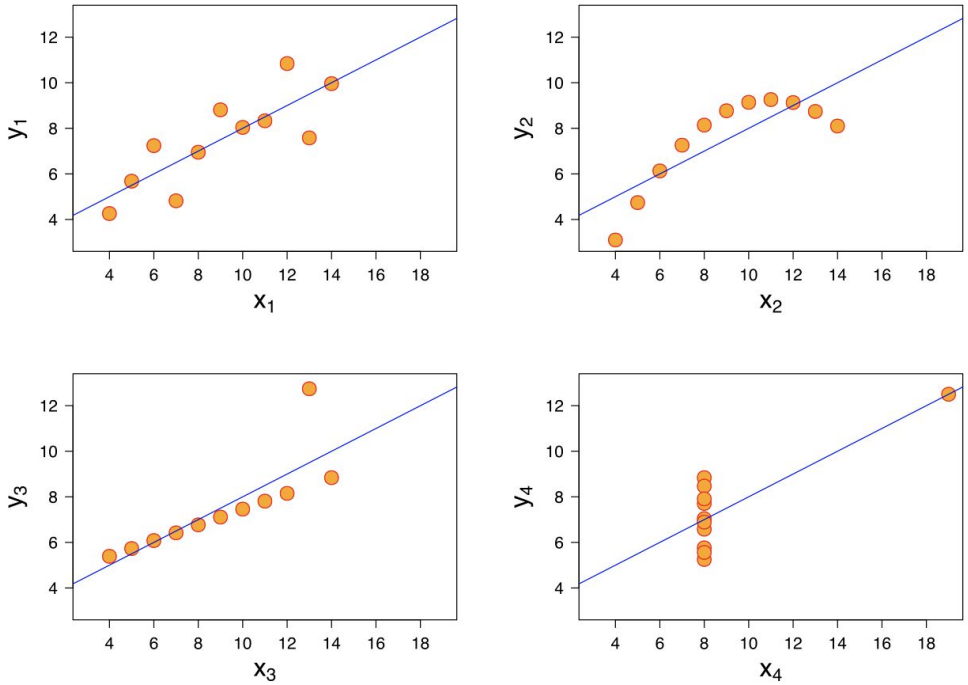
- We know the process of using a combination of statistics, tables, and graphs to better understand a dataset is **exploratory data analysis (EDA).**

 © 2020 General Assembly

Caveat: Anscombe's Quartet

Descriptive statistics are crucial. However, we know they don't always tell the whole story. This is **Anscombe's Quartet**, a famous set of four (x, y) datasets that all have the same:

- x-means, y-means
- x-standard deviations, y-standard deviations
- (x, y) correlations
- Lines of best fit



 © 2020 General Assembly

Data Vocab

What is “Data,” Anyway?

When I say “data,” you might immediately think “spreadsheet,” but this is wrong! While that is one representation of data, data is more synonymous with **information**. Data can come in many forms:

- Images
- Plain text
- HTML
- Scalar variables (x = 3)
- Yes, spreadsheets

 © 2020 General Assembly

Aspects of a DataFrame

Generally, in a data frame:

- One row represents one **observation** of data
  - Synonyms: sample, example
- One column represents one **variable**
  - Synonyms: field, feature, covariate

This is not always the case, but if it is, we can call the data **tidy**. We *love* tidy data. An example of *untidy* data would be a summarization of data, or a frequency table.

Variable Types

- **Quantitative / numeric** variables represent some quantity or magnitude of something.
  - Note not all numbers are quantitative:
  - When in doubt, ask “does the mean of this column make sense to find?”
- **Qualitative / categorical** variables represent categories, groups, or text.

 © 2020 General Assembly

Variable Types

- A subtype of categorical data is **ordinal data**. Ordinal data are categorical variables that have some ordering to them, but are not numeric.
  - For example:

- Some categorical data is **unordered**. Like:

- Another kind of variable is **datetime data**. In practice, you'll transform this to a number or a category to solve a problem.

 © 2020 General Assembly

Quantitative Subtypes

Quantitative variables can be

- **Discrete**: They take on **countable** sets of numbers
  - This often, but not always, means integers.
  - Countable does not necessarily mean finite!

- **Continuous**: They can take on *any* value on some interval.
  - Examples:
  - They only need to be *theoretically* continuous, even if not measured that way.
    - Example: Your height.

 © 2020 General Assembly

Reminder: Descriptive Statistics

- Descriptive statistics **summarize one or more variables** using a few numbers.
  - A **statistic** is a value computed from our data.

- This is often *really* important for communicating with other people about your work. We might want to understand or communicate:
  - The “center” of a variable.
  - How much a variable is spread out around that “center.”
  - How much two variables relate to each other.

 © 2020 General Assembly

Measures of Central Tendency

The Mean

The most well-known measure of **center** is the **mean** (or **average**, but I'll try to not use that word).

The mean is found by summing up all of the data and dividing by the number of observations we have.

The Mean

The most well-known measure of **center** is the **mean** (or **average**, but I'll try to not use that word).

The mean is found by summing up all of the data and dividing by the number of observations we have.



Yuck! This is words! Give me some math!

 © 2020 General Assembly

Some notation...

Given *n* observations of a variable, we'll denote it as a **vector** like this:

$$\mathbf{x} = (x_1, x_2, \ldots, x_n)$$

 © 2020 General Assembly

Some notation...

Given *n* observations of a variable, we'll denote it as a **vector** like this:

$$\mathbf{x} = (x_1, x_2, \ldots, x_n)$$



What's a "vector"?

 © 2020 General Assembly

The Sample Mean

So, given the data

$$\mathbf{x} = (x_1, x_2, \ldots, x_n)$$

The **sample mean** is then:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

 © 2020 General Assembly

Sensitive vs Resistant

We often call the sample mean a **sensitive** measure of center, since it's very sensitive to the presence of an outlier.

- What is the mean net worth of everyone on this call? (don't actually answer that)
- If Bill Gates joined this call, what would that number now become?
- Is that an "average" of all of us, really?

Can you think of a more outlier-**resistant** measure of center?

The Median

Sort all of your data, smallest to largest, and take the middle value. This is the **sample median** of your data.

Think about this in terms of the Bill Gates problem of the last slide. You'll see this statistic is more **resistant** (or **robust**) to outliers.

The Median

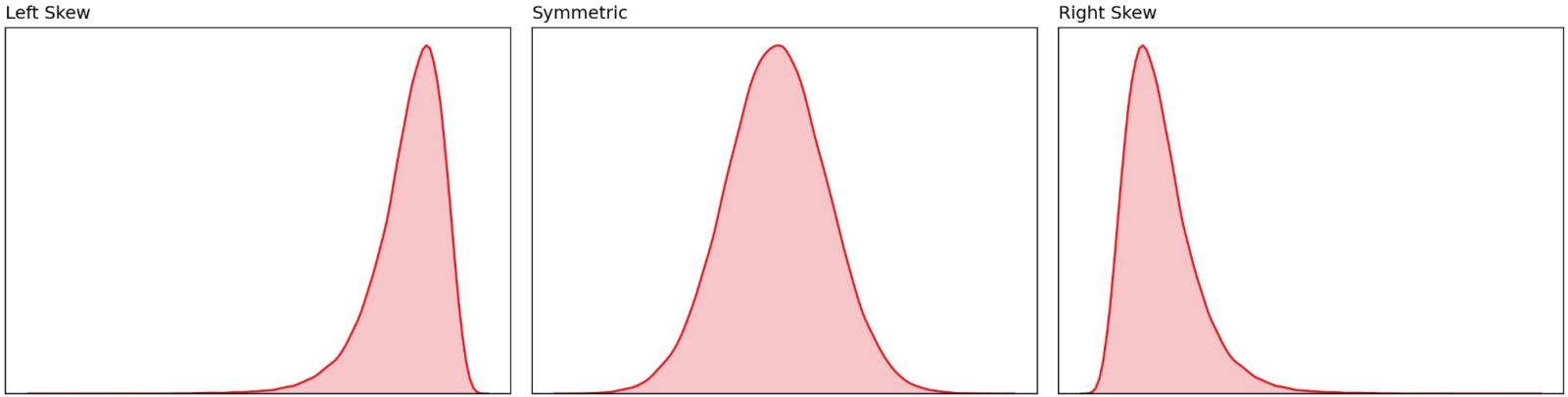
In the scope of our class, we don't have too many fancy symbols or notations for the median. But if you sorted all n of our observations, then the median would be...

- Observation $(n + 1) / 2$ if n is odd
- The mean of observation $n/2$ and $n/2 + 1$ if n is even

 © 2020 General Assembly

Aside: Skewness

The relationship between the mean and median can be used to get a quick ad hoc view of the **skewness** of our data:



What relationship between the mean and median could indicate right skewness?

 © 2020 General Assembly

The Mode

In grade school, you're often taught about the **mode** of a dataset. The mode is the most common value a variable can take.

The Mode

In grade school, you're often taught about the **mode** of a dataset. The mode is the most common value a variable can take.

THE MODE IS ABSOLUTELY WORTHLESS FOR QUANTITATIVE DATA, NO MATTER WHAT ANYONE TELLS YOU!

The Mode

In grade school, you're often taught about the **mode** of a dataset. The mode is the most common value a variable can take.

THE MODE IS ABSOLUTELY WORTHLESS FOR QUANTITATIVE DATA, NO MATTER WHAT ANYONE TELLS YOU!

There are only three times where the mode might be at all interesting:

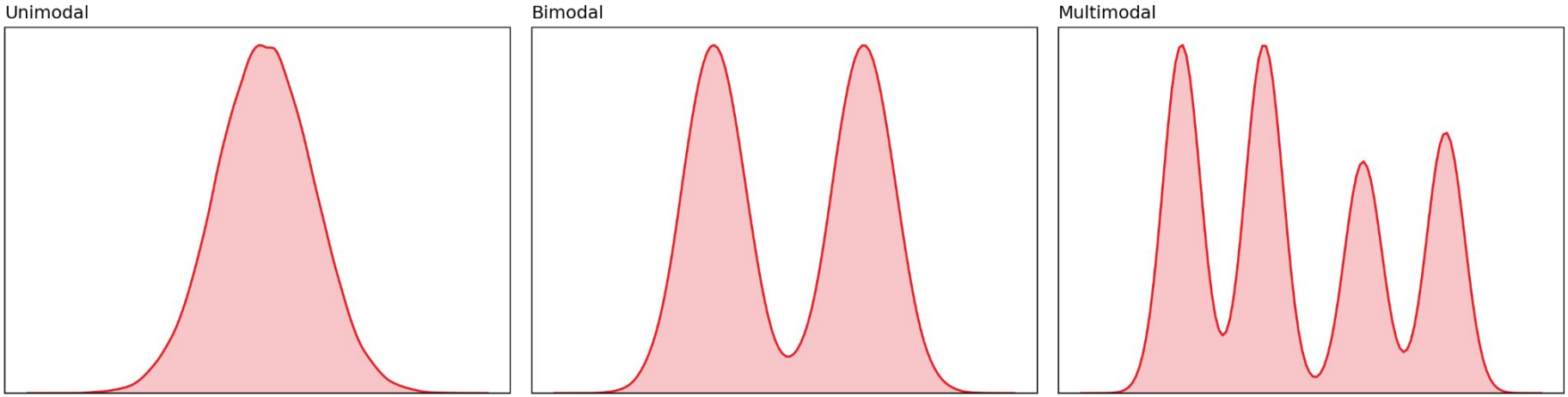
© 2020 General Assembly

Mode #1: Categorical Data

Sometimes people will refer to the most common category in a categorical column to be the **mode**. That's often interesting!

Mode #2: Modality

For quantitative data, we often care about the **modality** of our data



Mode #3: Exploratory Data Analysis

For data, the mode might be something interesting to explore during EDA.

Why might that be the case?

Aside: Exotic Measures of Center

The **trimmed mean** is the mean with some of the highest and lowest values removed

- eg: The 5% trimmed mean would remove the 2.5% lowest and 2.5% highest observations and take the mean of everything left.

The **winsorized mean** is the mean with some of the highest and lowest values replaced with a certain percentile

- eg: The 5% winsorized mean would transform the top 2.5% observations with the 97.5th percentile and the bottom 2.5% observations with the 2.5th percentile

Let's Try!



Measures of Spread (or Variability)

Measures of Spread

Measures of center don’t tell the whole story. They’re not complete unless we also include some measure of spread. That is, how the data *around the mean* are distributed.

You actually already know a few:

Min, Max, Range

The **minimum** and **maximum** value of a variable are technically measures of spread.

The **range** is the difference between the min and max. It's not as commonly used as you might think.

All three of these measurements are incredibly **sensitive** to outliers.

Sidebar: Percentiles

The *p*th percentile of a set of data is the number at which *p*% of the data lie below that number.

For example, the **median** is also the **50th percentile** of the data

The **minimum** and **maximum** would be the **0th** and **100th percentiles**, respectively.

 © 2020 General Assembly

Percentiles

We will often like to see the **25th and 75th percentiles**, which are denoted $\boldsymbol{Q_1}$ and $\boldsymbol{Q_3}$ (the first and third quartiles). These are are more **resistant** measures of spread, although not totally useful.

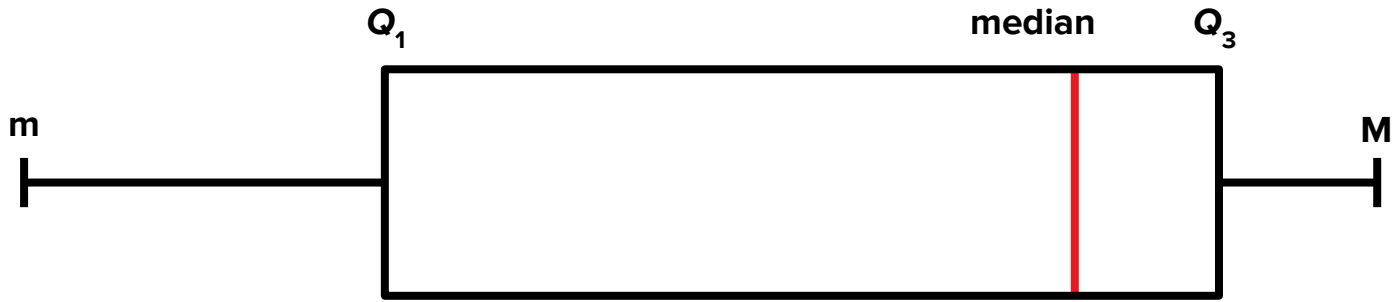
The difference of these two values is known as the **interquartile range (IQR):**

$$IQR = Q_3 - Q_1$$

 © 2020 General Assembly

The Five-Number Summary and Boxplots

The combination of the min, Q_1, median, Q_3, and max is known as the **five-number summary** of a dataset, and is mostly commonly used graphically to make up a **boxplot**:



 © 2020 General Assembly

The Variance

Let’s invent a measure of spread. In general, I want to know how far my data vary from their center (let’s say the mean). So, maybe I’m interested in:

The Variance

Let's invent a measure of spread. In general, I want to know how far my data vary from their center (let's say the mean). So, maybe I'm interested in:

$$x_i - \bar{x}$$

for each observation. This is a **deviation**.

 © 2020 General Assembly

The Variance

Each deviation represents how far a data point is from the mean! So let's take their average!

$$\frac{1}{n}\sum(x_i - \bar{x})$$

The Variance

Each deviation represents how far a data point is from the mean! So let's take their average!

$$\frac{1}{n}\sum(x_i - \bar{x})$$

Some of these deviations are positive and some are negative, however. Turns out, this always comes out to 0! So let's square each deviation to stop this from happening.

The Variance

We're close! Only one more issue...

$$s^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$$

The Variance

This is the **sample variance** for a quantitative variable. It represents on average, how far your data are from their mean... squared... over *n* - 1 instead of *n*...

$$s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$$

 © 2020 General Assembly

The Variance

The *n* - 1 thing is there for theory reasons beyond our scope, but it doesn't make a big difference for large *n*.

But the square thing is an issue. So let's take a square root!

$$s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$$

 © 2020 General Assembly

The Standard Deviation

This is the **sample standard deviation**, and it can be ***roughly*** interpreted as “the average distance each observation in your data are from the mean.” We’ll see this value a LOT, so let’s get a good feel for some real values.

$$s = \sqrt{\frac{1}{n-1}\sum(x_i - \bar{x})^2}$$

 © 2020 General Assembly

SD Example 1:

You're a traffic engineer monitoring a busy highway. You record the speeds of 50 cars and compute a sample mean of 67 mph. Which of the following is a realistic guess for a standard deviation of this data:

A. 40 mph
B. -8 mph
C. 7 mph
D. 100 mph

 © 2020 General Assembly

SD Example 2:

You agree to play a betting game in which a process generates a random number from a bell-shaped, unimodal, symmetric distribution. If the number's positive, you win that much, but if it's negative, you lose that much. Denote the pair (mean, sd). Which game would you prefer to play?

A. (10, 3)
B. (100, 200)
C. (50, 50)

(There is no right answer, just discuss!)

Let's Try!



 © 2020 General Assembly

Standardizing Data with Z-scores

Standardization and the Z-score

A major difficulty in dealing with so many different types of data is their **scale** or **units**. If you were to pick a random value from a column of numeric data, can you tell if it's high or low at a glance? With unknown scale?

e.g.: I have data for marathon run times. I'm not a runner. I have no idea what constitutes a good run. I pick a random value of 4hr 53min. Is that good?

Z-scores

Start by finding the sample mean and sample SD. Take that observation, subtract the mean, and then divide by the SD. You now have that observation's **z-score**:

$$z_i = \frac{x_i - \bar{x}}{s}$$

Z-scores

Note that the z-score is 0 if and only if that observation is the same as the mean. Z-scores can be a useful trick to get a quick view of the magnitude of an observation:

- A z-score of near 0 indicates something is average.
- A positive z-score means something is above average
  - We consider a z-score of 2 to be “pretty high”
- A negative z-score means something is below average
  - We consider a z-score of -2 to be “pretty low”

 © 2020 General Assembly

Standardization

Turning an entire column of data into their corresponding z-scores is known as **standardizing** that variable, since we've put that variable on the "standard" scale of having **mean 0 and standard deviation 1 with no units** instead of minutes, mph, miles, etc.

MISCONCEPTION: Beginner statistics students often erroneously believe that standardizing a column makes it normal (bell-shaped). This is false! It doesn't change the shape of the data at all! However, if our data was already normal, we can make a lot of really cool statements, but more on that later.

 © 2020 General Assembly

Let's Try!



Covariance and Correlation

(time permitting)

Covariance and Correlation

How are two quantitative variables related to one another? Do they vary together? If one rises, will the other rise? The metric for this is called **covariance** but we won't really talk about it.

It's mostly used for doing certain theory calculations — like principal component analysis (PCA).

Correlation

However, it's possible to scale the covariance to be between -1 and 1. This is called the **correlation [coefficient]**. We *love* this number.

Caveat: The correlation between X and Y only describes the **linear** relationship between those variables. It's totally possible to have an obvious non-linear relationship between X and Y and a 0 correlation.

Reminder...

CORRELATION DOES NOT IMPLY CAUSATION!

Just because X and Y have high correlation, does not mean X causes Y or even that Y causes X!

Sample Correlation

Don’t stare at this too long:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

 © 2020 General Assembly

Covariance and correlation: generally

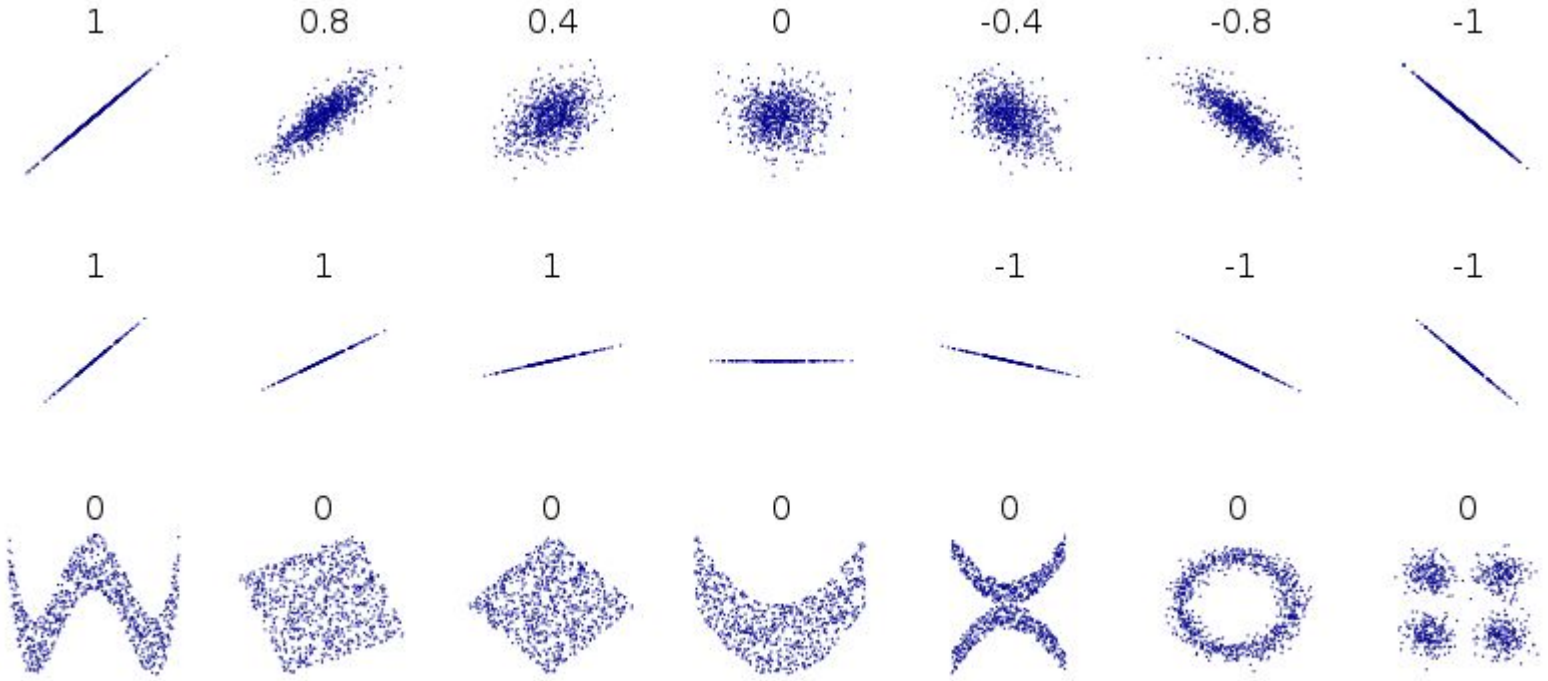


Image source: Wikipedia

Guessing Game

http://www.istics.net/Correlations/

Let's Try!



 © 2020 General Assembly